

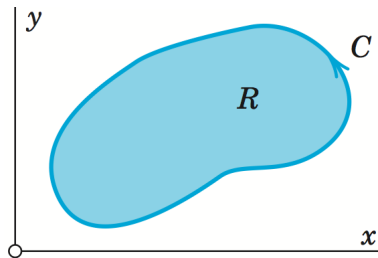
Topic: Green's Theorem

I. Introduction

In this section we will on line integrals over closed curves in the x, y -plane.

Notation: When representing a line integral over a closed curve, we will use the notation

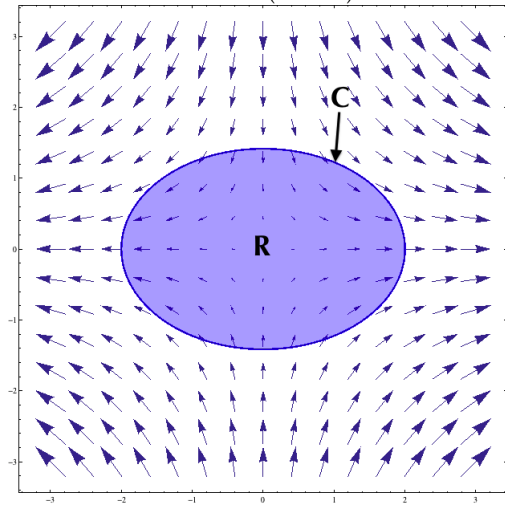
$$\oint_C \mathbf{F} \cdot d\mathbf{r} \quad \text{instead of} \quad \int_C \mathbf{F} \cdot d\mathbf{r}$$



Boundary of R is the Closed Curve C

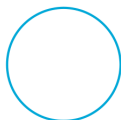
Note: Any planar closed curve can be thought of as the boundary of a region, R , in the plane.

F – Vector Field (Arrows)



For the rest of this section, the curves that we will consider have the property of being **simple, closed curves**.

Defn: A curve is said to be **simple** if it never **crosses itself**.



Simple



Simple



Not simple

Q: Is there a relationship between the region, R , the boundary, C , and vector field, F ?

A:

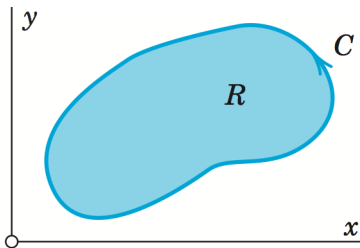
II. Green's Theorem

Thm: Let C is a simple closed curve bounding a planar region R . If $\mathbf{F}(x, y) = [M, N]$ is a vector field, then

$$\text{(General Form)} \quad \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R (\text{curl } \mathbf{F}) \cdot \mathbf{k} \, dA$$

Since $\mathbf{F} = [M, N]$, a quick calculation shows: $(\text{curl } \mathbf{F}) \cdot \mathbf{k} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$, we have:

$$\text{(Component Form)} \quad \oint_C (M \, dx + N \, dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$



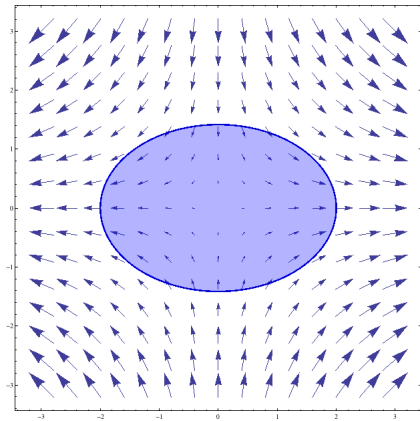
III. Forms of Green's Thm

Tangential Form:

$$\oint_C \mathbf{F} \cdot \mathbf{T} \, ds = \iint_R (\text{curl } \mathbf{F}) \cdot \mathbf{k} \, dA$$

$\mathbf{T}$ is the unit tangent vector along C .

Quantifies the tendency of the vector field to act in the direction of the boundary (circulation).

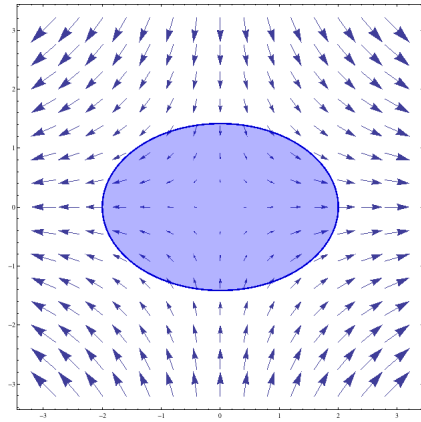


Normal Form:

$$\oint_C \mathbf{F} \cdot \mathbf{N} \, ds = \iint_R \text{div } \mathbf{F} \, dA$$

$\mathbf{N}$ is the unit normal vector along C .

Quantifies the tendency of the vector field to act perpendicularly to the boundary (flux).



Before verifying these let's summarize some notation:

$$(i) \quad \mathbf{r}'(t) = [x'(t), y'(t)] \quad (ii) \quad \mathbf{T} = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \quad (iii) \quad \mathbf{N} = \mathbf{T} \times \mathbf{k} \quad (iv) \quad d\mathbf{r} = \mathbf{r}'(t) dt \quad (v) \quad ds = |\mathbf{r}'(t)| dt$$

Tangential Form:
$$\oint_C \mathbf{F} \cdot \mathbf{T} ds = \iint_R (\text{curl } \mathbf{F}) \cdot \mathbf{k} dA$$

$$\oint_C \mathbf{F} \cdot \mathbf{T} ds = \oint_C \mathbf{F} \cdot \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} (|\mathbf{r}'(t)| dt) = \oint_C \mathbf{F} \cdot \mathbf{r}'(t) dt = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R (\text{curl } \mathbf{F}) \cdot \mathbf{k} dA \quad \checkmark$$

Normal Form:
$$\oint_C \mathbf{F} \cdot \mathbf{N} ds = \iint_R \text{div } \mathbf{F} dA$$

$$\begin{aligned} (*) \quad \mathbf{N} ds &= (\mathbf{T} \times \mathbf{k}) ds = \left(\frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \times \mathbf{k} \right) (|\mathbf{r}'(t)| dt) = \frac{1}{|\mathbf{r}'(t)|} (\mathbf{r}'(t) \times \mathbf{k}) (|\mathbf{r}'(t)| dt) \\ &= (\mathbf{r}'(t) \times \mathbf{k}) dt \\ &= ([x'(t), y'(t), 0] \times [0, 0, 1]) dt \\ &= [y'(t), -x'(t), 0] dt \\ &= \left[\frac{dy}{dt}, -\frac{dx}{dt}, 0 \right] dt \\ &= [dy, -dx, 0] \\ &= [dy, -dx] \end{aligned}$$

and so, by (*), we have ...

$$\begin{aligned}\oint_C \mathbf{F} \cdot \mathbf{N} \, ds &= \oint_C \mathbf{F} \cdot (dy, -dx) = \oint_C [M, N] \cdot [dy, -dx] \\&= \oint_C M \, dy - N \, dx \\&= \oint_C -N \, dx + M \, dy \\&= \iint_R (M)_x - (-N)_y \, dA \\&= \iint_R M_x + N_y \, dA \\&= \iint_R \operatorname{div} \mathbf{F} \, dA \quad \checkmark\end{aligned}$$

Ex: Let $\mathbf{F} = [x^2 + 4y, x + y^2]$ and let C be a square bounded by: $x = 0$, $x = 1$, $y = 0$, and $y = 1$,

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

To compute this directly we would have to compute a line integral for C_1 , C_2 , C_3 , and C_4

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_{C_1} \mathbf{F} \cdot d\mathbf{r} + \oint_{C_2} \mathbf{F} \cdot d\mathbf{r} + \oint_{C_3} \mathbf{F} \cdot d\mathbf{r} + \oint_{C_4} \mathbf{F} \cdot d\mathbf{r}$$

and we would need to parameterize (find $\mathbf{r}(t)$) all 4 curves

Alternatively, we could apply Green's Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

$$\frac{\partial N}{\partial x} = 1 \quad \text{and} \quad \frac{\partial M}{\partial y} = 4$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_R (1 - 4) dA = -3 \iint_R dA = -3 (\text{Area of } R) = -3$$

Ex: Let $\mathbf{F} = [16xy + 3yx^2, x^3 + 2y]$ and let C be a triangle bounded by: $x = 3$, $y = 0$, and $y = x$

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

First compute: $N_x = (x^3 + 2y)_x = 3x^2$

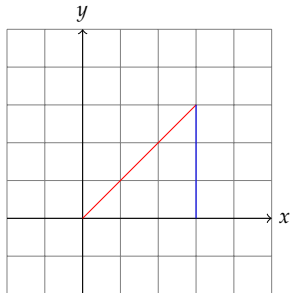
and

$M_y = (16xy + 3yx^2)_y = 16x + 3x^2$

and apply Green's Thm:

$$\begin{aligned}\oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_R (3x^2) - (16x + 3x^2) dA \\&= \iint_R -16x dA \\&= \int_0^3 \int_0^x -16x dy dx \\&= \int_0^3 -16xy \Big|_{y=0}^{y=x} dx \\&= \int_0^3 -16x^2 dx = -\frac{16}{3}x^3 \Big|_0^3 = -\frac{16}{3}(3)^3 = -144\end{aligned}$$

Plot the region bounded by C :



Ex: Let $\mathbf{F} = [xy, y^2]$ and let R be the region in the first quadrant bounded by: $y = x^2$, and $y = x$

Compute $\oint_C \mathbf{F} \cdot \mathbf{T} ds$ and $\oint_C \mathbf{F} \cdot \mathbf{N} ds$

Ex: Let $\mathbf{F} = \nabla(x^2 \sin y)$ and let R be the region bounded by: $x = 1$, $y = 0$, and $y = \sqrt{x}$

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

Summary: Line Integral over Vector-Valued Functions

We have 3 main methods for computing line integrals over a vector-valued function $\mathbf{F}$:

- (1) Directly $\left[\text{i.e. find } \mathbf{r}(t) \text{ and compute } \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \right]$
- (2) Fundamental Thm of Line Integrals $\left[\text{i.e. find } f \text{ s.t. } \nabla f = \mathbf{F} \implies \int_A^B \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A) \right]$
- (3) Green's Thm $\left[\text{i.e. apply } \int_C \mathbf{F} \cdot d\mathbf{r} = \iint_R (N_x - M_y) dA \right]$

C closed?	F Conservative? (curl $\mathbf{F} = 0$?)	Use Method	$\mathbf{r}$ needed?
Yes	Yes	$\int_C \mathbf{F} \cdot d\mathbf{r} = 0$	No
	No	(3)	No
No	Yes	(2)	No
	No	(1)	Yes

Note: Method (1) can always be applied to any line integral